

RYGGE MOSS, Norway

WMO: 014940

Lat: 59.379N

Lon: 10.786E

Elev: 174

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.8	(c) 8.6	(d) 0.4	(e) 5.6	(f) 6.3	(g) 4.7	(h) 7.1	(i) 11.8	(j) 25.5	(k) 40.7	(l) 23.2	(m) 40.2	(n) 3.2	(o) 0	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.8	(c) 78.7	(d) 63.3	(e) 75.2	(f) 62.1	(g) 71.9	(h) 61.0	(i) 66.3	(j) 73.7	(k) 64.7	(l) 71.6	(m) 63.1	(n) 69.1	(o) 7.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.5	68.6	62.3	84.5	67.0	60.7	79.7	65.7	31.0	73.5	29.8	71.3	28.6	69.0	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.2	(b) 18.8	(c) 16.5	(e) -0.8	(f) 82.7	(g) 6.5	(h) 4.3	(i) -5.4	(j) 85.8	(k) -9.3	(l) 88.3	(m) -12.9	(n) 90.7	(o) -17.7	(p) 93.8		
(4) 21.2	(5) 18.8	(6) 16.5	(7) -0.8	(8) 82.7	(9) 6.5	(10) 4.3	(11) -5.4	(12) 85.8	(13) -9.3	(14) 88.3	(15) -12.9	(16) 90.7	(17) -17.7	(18) 93.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	27.8	28.8	34.2	42.6	51.4	58.3	63.3	61.5	54.5	45.1	36.8	30.4		
	DBStd	14.21	9.27	8.93	6.91	5.68	6.16	4.53	4.37	4.30	4.98	6.25	7.29	9.78		
	HDD50	3256	687	594	491	232	54	1	0	18	174	397	608			
	HDD65	7491	1152	1014	955	672	424	209	87	128	315	616	846	1073		
	CDD50	1300	0	0	0	10	97	249	411	356	153	23	1	0		
	CDD65	60	0	0	0	0	2	7	33	18	0	0	0	0		
	CDH74	406	0	0	0	0	23	65	236	82	0	0	0	0		
	CDH80	54	0	0	0	0	1	6	43	4	0	0	0	0		
	WSAvg	7.6	7.3	7.6	7.6	7.8	7.9	8.1	7.5	7.2	7.6	7.5	7.6	7.2		
	PrecAvg	35.9	3.1	2.3	1.8	2.2	2.3	2.9	3.1	3.9	3.3	4.2	3.9	3.1		
(15) Precipitation	PrecMax	50.0	6.9	5.7	5.5	4.6	4.3	5.7	6.2	7.3	8.4	10.7	12.0	6.3		
	PrecMin	27.3	0.5	0.6	0.0	0.4	0.2	0.3	0.2	1.6	1.1	0.6	1.0	0.7		
	PrecStd	5.9	1.6	1.3	1.2	1.1	1.1	1.2	1.3	1.5	1.7	2.3	2.4	1.7		
	PrecStd	5.9	1.6	1.3	1.2	1.1	1.1	1.2	1.3	1.5	1.7	2.3	2.4	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	46.1	47.5	56.9	66.6	77.5	80.3	84.4	80.2	70.3	59.9	53.6	49.0			
	0.4% MCWB	42.8	40.7	45.8	52.6	60.8	62.3	66.1	63.7	59.1	53.9	50.4	47.1			
	2% DB	43.4	43.3	50.3	61.0	71.5	75.4	80.3	76.4	66.6	57.4	51.4	46.8			
	2% MCWB	41.1	39.5	42.2	49.5	56.6	59.8	64.7	63.9	57.9	53.3	49.3	45.2			
	5% DB	41.4	41.4	46.3	56.4	66.6	71.3	75.9	73.2	64.5	55.7	49.2	45.0			
	5% MCWB	39.5	38.8	39.2	46.7	54.6	58.3	63.0	62.8	57.4	52.7	47.3	43.1			
	10% DB	39.5	39.9	43.0	52.6	62.6	67.7	72.6	69.8	62.4	54.1	47.1	42.9			
	10% MCWB	37.9	37.8	38.3	44.5	52.4	56.7	62.1	61.4	56.7	51.5	45.3	41.4			
	0.4% WB	44.2	43.0	46.8	55.0	62.8	65.0	68.7	68.2	63.3	56.6	52.2	48.0			
	0.4% MCDB	44.9	44.4	55.3	63.5	71.6	73.6	78.0	73.5	66.1	57.9	52.7	48.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	41.9	41.1	43.1	50.4	59.2	62.4	66.8	66.1	61.1	55.1	49.6	45.8			
	2% MCDB	42.6	42.2	47.8	58.3	67.1	71.4	75.7	71.7	63.9	56.3	50.5	46.6			
	5% WB	40.0	39.7	41.1	48.2	56.5	60.4	65.2	64.5	59.4	53.8	47.6	43.5			
	5% MCDB	40.9	40.9	44.1	54.2	63.6	67.4	73.0	70.5	62.5	55.1	48.7	44.5			
	10% WB	38.2	38.0	39.4	46.1	54.2	58.7	63.6	62.8	57.8	52.2	45.6	41.4			
	10% MCDB	39.3	39.5	42.0	50.7	61.0	65.0	70.2	68.1	61.0	53.9	46.9	42.6			
(35) Mean Daily Temperature Range	MDBR	8.9	10.0	12.4	14.2	14.9	13.9	13.8	13.5	12.7	10.4	8.2	8.6			
	5% DB	8.9	9.8	16.9	21.1	22.0	20.6	20.8	18.1	15.2	10.7	9.8	7.9			
	5% WB	8.5	7.7	10.7	11.7	10.6	9.4	9.0	8.6	9.1	8.3	9.7	8.1			
	MCDBR	7.8	8.4	14.2	17.6	18.0	16.2	16.9	15.2	11.7	8.8	8.9	7.9			
	MCWBR	8.1	7.4	9.8	10.6	9.8	8.9	8.6	8.4	8.5	8.1	9.5	8.4			
	MCWBR	8.1	7.4	9.8	10.6	9.8	8.9	8.6	8.4	8.5	8.1	9.5	8.4			
(40) Clear-Sky Solar Irradiance	taub	0.278	0.314	0.335	0.374	0.369	0.357	0.378	0.373	0.343	0.331	0.295	0.253			
	taud	2.234	2.317	2.366	2.322	2.411	2.463	2.425	2.434	2.506	2.440	2.294	2.154			
	Ebn at Noon	167	222	254	262	273	278	268	261	252	215	159	130			
	Edh at Noon	14	21	27	34	33	32	32	30	24	19	13	10			
(44) All-Sky Solar Radiation	RadAvg	97	282	760	1207	1602	1832	1638	1290	859	380	129	66			
	RadStd	18	51	81	167	159	122	200	91	95	70	33	7			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page